

From Air to Log-Mel

A study guide, stitched from the manim-concepts road

manim-concepts

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How to read this guide

The objective is to read a log-mel spectrogram with understanding — the picture a speech model is handed in place of sound — knowing what every axis, every row and every shade of it measures, and what was thrown away to make it. The road has four stations, one per video series of the repo’s signal-processing topic:



This guide grows with the road. *The spectrum* is written: one column of the picture, built as a bank of detectors on samples you can count. The dashed stations are planned and not yet written — *convolution* (the sliding weighted sum, and the theorem that explains leakage), *windowing, the short-time transform and the spectrogram* (the columns side by side), and *mel* (the rows regrouped the way the ear groups them, with a log on top). Each is written when it is asked for, and with it comes a transition page that says why the next station needs exactly what the last one built.

The first chapter leans on three chapters of this repo’s other guide, *The CTC Algorithm*: random variables (the weighted sum), logarithms (multiplying is adding, and $\log_{10} 2$) and the counting rules ($\binom{n}{k}$). Each use restates what it needs, so the chapter reads on its own. Practice problems close every chapter — sized to double as a diagnostic before you study digital audio — and worked solutions live in the separate solutions manual, keyed by the same problem numbers. Worked numbers are spliced from the repo’s verification anchors — for instance, the chapter’s two-tone mix reads 0, 4, 2, 0, 0 on its five detectors — never retyped.

Chapter 1

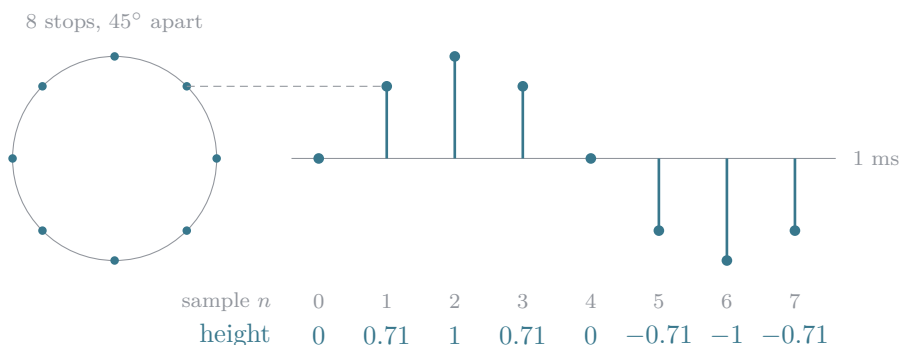
The spectrum: a bank of detectors

A speech model never hears a sound. It reads a picture of one — a spectrogram — and every column of that picture is the object this chapter builds: a spectrum. The usual introduction makes the spectrum a new kind of thing, reached through a new kind of number. This chapter makes it an old one. Eight samples go in; five detectors listen to the same eight samples, each at its own fixed frequency; and each detector does nothing but the multiply-and-add that the *random-variables* chapter built for expectation. The spectrum is a column of weighted sums, read off a bank of detectors — the formulation Steven W. Smith calls the real DFT [31]. All the arithmetic runs on one grid: $N = 8$ samples at a sample rate of $sr = 8000$ per second.

1.1 Pressure into numbers

Sound is air pressure changing over time, and the simplest sound is a pure tone. Picture it as a point going round a circle at a steady speed; the microphone reads the point's *height*. One lap per millisecond is a thousand laps a second: a 1000 Hz tone. A computer cannot keep the whole motion. It keeps the height at regular instants — 8000 of them a second, so 8 in this one millisecond, one every 45° of the lap. The 8 stops on the circle *are* the 8 samples:

$$x[n] = \sin(2\pi n/8), \quad n = 0, \dots, 7 : \\ 0, 0.71, 1, 0.71, 0, -0.71, -1, -0.71.$$



The samples are drawn as stems because nothing between them is stored. A staircase — each value held until the next — draws information the recording does not contain; as Montgomery’s demonstration puts it, the stairsteps aren’t really there [19]. Read the same point sideways and you get its *shadow*: the same motion a quarter-turn ahead,

$$1, 0.71, 0, -0.71, -1, -0.71, 0, 0.71$$

— two readings of one rotation. The height is the sine and the shadow is the cosine, and the next section needs both.

Every digital recording starts here: 8000 samples a second for telephone speech, 16 000 for speech models [13]. One thing is planted for the last section: this tone gets 8 samples per lap; a faster tone gets fewer, and needs at least two. And real sound is never one tone. Here are 8 samples of a mix,

$$0.3536, 0.3536, 0.6464, 1.0607, 0.3536, -1.0607, -1.3536, -0.3536,$$

and the chapter’s question: which tones are hiding in them?

1.2 The probe

To ask whether a 2000 Hz tone is present, build one: a *probe* is a known tone that fits a whole number of laps into the window, and the 2-lap sine probe happens to be all integers at the 8 stops — 0, 1, 0, -1, 0, 1, 0, -1. Multiply the samples by the probe, stop by stop, and add. That is the weighted sum $\sum_x x \cdot P(X = x)$ of the *random-variables* chapter, with two differences stated plainly: these weights can be negative, and they add to 0. They are a pattern to match, not a probability.

2-lap tone	0	1	0	-1	0	1	0	-1	
2-lap probe	0	1	0	-1	0	1	0	-1	
products	0	1	0	1	0	1	0	1	sum = 4
1-lap tone	0	0.71	1	0.71	0	-0.71	-1	-0.71	
3-lap probe	0	0.71	-1	0.71	0	-0.71	1	-0.71	
products	0	0.5	-1	0.5	0	0.5	-1	0.5	sum = 0

A *match* first: the 2-lap tone against the 2-lap probe. Every product is a square, so nothing can cancel: $0 + 1 + 0 + 1 + 0 + 1 + 0 + 1 = 4$. Then a *mismatch*: a 1-lap tone against the 3-lap probe. The products take both signs — four halves up, two wholes down — and the sum is exactly 0. This detector hears nothing. It is Smith's correlation picture, rebuilt small enough to count [27].

One probe is not enough, though. Take a louder 2000 Hz tone whose samples run 3, 4, -3, -4 and repeat. The sine probe reads 16. Start the *same tone* one sample later and it reads 12: the detector has been fooled by when the tone starts [25]. The repair is the shadow. Run the 2-lap cosine probe as well and keep the *pair*,

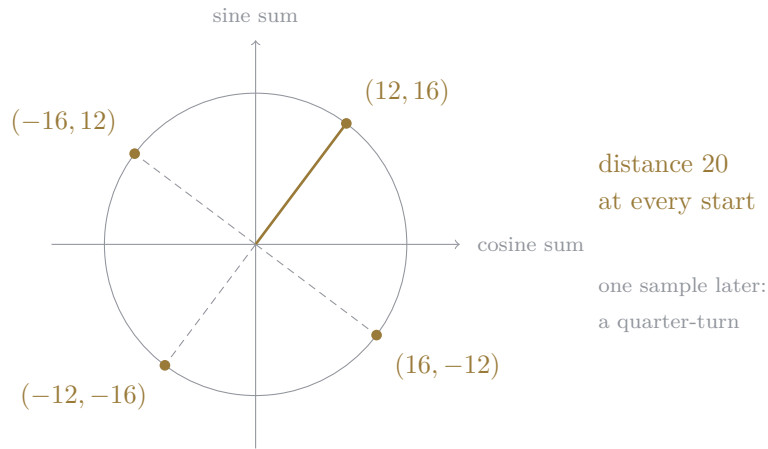
$$a_k = \sum_{n=0}^7 x[n] \cos(2\pi kn/8), \quad b_k = \sum_{n=0}^7 x[n] \sin(2\pi kn/8),$$

$$\text{reading} = \sqrt{a_k^2 + b_k^2},$$

both sums plus-signed. At the four starts the pair is

$$(12, 16), (-16, 12), (-12, -16), (16, -12) :$$

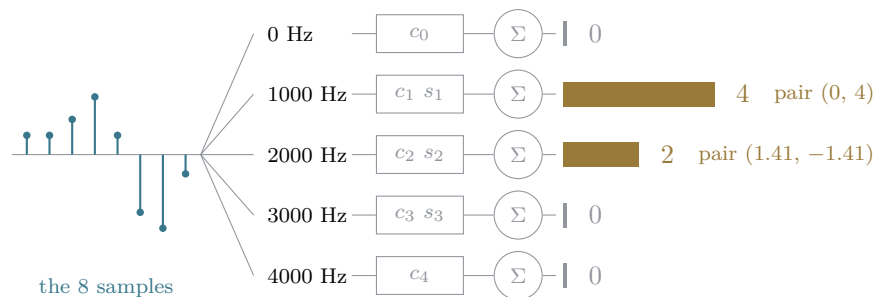
one sample later is a quarter-turn of the pair, and its distance from the origin is 20 every time. The sine probe alone saw only the height of that turning point — 16, 12, -16, -12.



The reading ignores when the tone starts — true for every starting angle, shown here for the four the grid offers. And it measures something the samples never show: $20 \div 4 = 5$ is the tone’s amplitude (the 4 is the match sum above), while the largest sample is only 4. One detector, then, is a pair of probes: multiply, add, take the distance. Every spectrum plot is this measurement repeated at each frequency.

1.3 The bank of detectors

Build one detector for each whole number of laps the window can hold — 0, 1, 2, 3, 4 — and label the rows 0, 1000, 2000, 3000 and 4000 Hz *before any sound arrives*. A row is an address, not a discovery; what the literature calls a bin is a detector’s reading at a fixed address, never “the frequency found in the signal” [11]. Feed the same 8 samples to all five. A 2000 Hz tone lights row 2 alone, and it reads 4. A 3000 Hz tone started somewhere else entirely — a cosine this time — lights row 3 alone, again 4. Then the mix from the first section, which is Lyons’ worked example [17] (the signal is his; the numbers here are computed): the five readings are 0, 4, 2, 0, 0.



Two rows answer. Row 1's pair is (0, 4) and row 2's is (1.41, -1.41) — the half-size tone started at 135° , so neither of its probes alone tells the truth, and its bar is the pair's distance, 2: the previous section's lesson inside the bank. Reading $\div 4$ gives the amplitudes, 1 and 0.5. Turn the column of bars on its side and it is the plot everyone calls *the spectrum* — no new object, a column of weighted sums of the same samples. The figure keeps the samples beside the bank on purpose: students who are shown spectra without their waveforms reliably fail to connect the two pictures [33].

In code the bank already exists. `np.fft.rfft` returns these five rows, each pair packed into one complex number — a device this road has not built, so take it as a pointer: its real part is the cosine sum and its imaginary part is *minus* the sine sum, $X[k] = a_k - i b_k$ [5]. Row 2 of the mix, (1.41, -1.41) here, is numpy's $1.41 + 1.41i$. The reading — the distance — is the same either way.

1.4 No double counting

Why did the mix split so cleanly into 4 and 2, with nothing spilled into the other rows? Two facts. First, each probe's sum on a mix is the sum of its sums on the parts: a weighted sum of a sum is the sum of the weighted sums, the same one-line reason the *random-variables* chapter gave for $E[X + Y] = E[X] + E[Y]$. (It is the *pair* that adds, cosine sum to cosine sum and sine sum to sine sum; the distance is taken last, and distances do not add — the sums 12 and 16 made 20.) So it is enough to ask what each probe sums to on every other probe. Second, the answer: nothing.

	c_0	c_1	s_1	c_2	s_2	c_3	s_3	c_4	
c_0	8	0	0	0	0	0	0	0	
c_1	0	4	0	0	0	0	0	0	$C(8, 2) = 28$ different pairs
s_1	0	0	4	0	0	0	0	0	all 28 read exactly 0
c_2	0	0	0	4	0	0	0	0	
s_2	0	0	0	0	4	0	0	0	c : cosine probe, s : sine probe
c_3	0	0	0	0	0	4	0	0	the number: laps per window
s_3	0	0	0	0	0	0	4	0	
c_4	0	0	0	0	0	0	0	8	

There are 8 probes — a cosine for each of the five rows, a sine for each of the middle three — and so 64 multiply-and-add sums. The diagonal, each probe against itself, reads 8, 4, 4, 4, 4, 4, 4, 8. Off the diagonal there are $\binom{8}{2} = 28$ different pairs, counted the way the *counting-rules* chapter counts them, and every one reads exactly 0. That is *computed*, *not proved*: an exhaustive check at $N = 8$. The reason it holds for every N needs a tool this road has not built — Euler’s formula — and waits for it [9]. The consequence needs no waiting: each detector is exactly deaf to the other detectors’ tones, so no bar can borrow from another.

Why 4 on the diagonal? Squares never cancel, and at every stop height² + shadow² = 1 — it is the circle’s radius, by Pythagoras. The squared sine probe and the squared cosine probe run

$$\begin{array}{l} 0, \frac{1}{2}, 1, \frac{1}{2}, 0, \frac{1}{2}, 1, \frac{1}{2} \\ 1, \frac{1}{2}, 0, \frac{1}{2}, 1, \frac{1}{2}, 0, \frac{1}{2} \end{array}$$

— the same values in a different order, summing to 1 at each stop. Eight stops share out 8 between two probes that take the same values — 4 each [8].

Why 8 at the two ends? At 0 laps and at 4 laps the sine probe is all zeros — $\sin 0$ and $\sin(\pi n)$ vanish at every stop [28] — so the cosine has no partner and keeps the whole 8. The end rows are therefore different in two ways, stated once: they read $\div 8$, not $\div 4$ [30]; and with only one probe they *do* care when the tone starts. A full-size tone on an end row reads 8, 6.9282, 5.6569, 0 as its start moves through 0° , 30° , 45° and 90° . This is why every worked tone in this chapter lives on rows 1 to 3.

The bookkeeping closes the argument: $1 + 2 \cdot 3 + 1 = 8$ readings from 8 samples [29]. Nothing is lost, which is why the transform can be undone — the 8 readings rebuild the 8 samples. That is a promise, not something shown here. And one warning rides along: each detector is deaf at the *other*

detectors' frequencies — and only there [7, 16]. The next section meets a tone that sits somewhere else.

1.5 What sets the spacing

What decides where the detectors listen? Probe k fits k laps into the window, and the window lasts $N \div sr$ seconds — here $8 \div 8000$ s, one millisecond. So k laps per millisecond is $k \times 1000$ Hz, and in general

$$f_k = k \cdot \frac{sr}{N}, \quad \text{spacing} = \frac{sr}{N} = \frac{1}{\text{duration}}.$$

The spacing between detectors is one over how long the window listens. Three banks, side by side, separate the two things a designer can buy:

$sr = 8000, N = 8$	1 ms, 1000 Hz apart, 5 rows	rows to 4000 Hz
$sr = 16\,000, N = 16$	1 ms, 1000 Hz apart, 9 rows	rows to 8000 Hz
$sr = 8000, N = 16$	2 ms, 500 Hz apart, 9 rows	rows to 4000 Hz

Sampling twice as fast for the same millisecond leaves the rows 1000 Hz apart — the faster rate bought *reach*, not detail. Listening twice as long put them 500 Hz apart: only duration buys spacing. The row count is $N/2 + 1$ in every case.

This is a speech model's front end. Whisper takes 400 samples at 16 000 Hz — a window of 25 ms — which is 201 rows, 40 Hz apart, up to 8000 Hz [24, 21]. And the spacing has a cost, which this chapter shows and does not explain. A 1500 Hz sine starting at 0 fits one and a half laps into the window — its 8 samples show it: not a whole number of laps — so it sits between two rows of the small bank, and *every* row answers — on rows 0 to 4,

$$1.50, 2.85, 2.41, 0.85, 0.67.$$

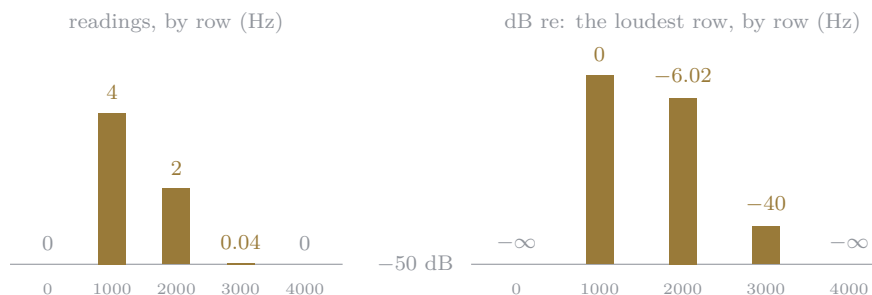
That smear is *leakage*, and it is the windowing chapter's subject — the one computed tone in this chapter that does not sit on a detector's frequency (the next section's trumpet note is the other, measured rather than computed). Two more pointers along the same road: slide the window along the recording and each detector's output becomes a signal in its own right, which is when the bank earns the name *filterbank* [12]; and a *mel* filterbank will regroup Whisper's 201 rows into 80.

1.6 The spectrum in decibels

Add a third tone to the mix, a hundred times quieter than the first: amplitude 0.01 at 3000 Hz. The readings are 4, 2, 0.04, and on ordinary bars the third tone is invisible. The cure is the counting strip of the *logarithms* chapter. There is one definition: a level in decibels is ten times the common logarithm of a *power* ratio [18]. Power goes as amplitude squared, and the law $\log(xy) = \log x + \log y$ — multiplying is adding — turns the square into a factor of 2 out front [6, 20]:

$$L = 10 \log_{10} \frac{P}{P_0} = 20 \log_{10} \frac{A}{A_0} \text{ dB.}$$

A decibel figure means nothing without its reference; here it is the loudest row. That row is ratio 1, and $\log 1 = 0$ dB. Half the reading is $20 \log_{10} \frac{1}{2} = -20 \times 0.301 = -6.02$ dB, on the logarithm chapter's own value of $\log_{10} 2$. A hundredth is two tenfoldings down: -40 dB — and now the third tone has a bar.



The empty rows are the boundary reading the logarithm chapter flagged: $\log 0 = -\infty$. So 0 dB means *equal to the reference* — it is never silence; silence is $-\infty$ dB. Real systems therefore pick a floor. Whisper keeps 8 decades of power below its loudest value — 80 dB — and clamps the rest [21]. Two conversions are worth carrying: doubling an amplitude is 6.0206 dB, doubling a power 3.0103 dB.

A real note shows what the axis is for. Take 4096 samples of a trumpet at 22 050 per second: 2049 detectors, 5.3833 Hz apart. Its first five harmonics, in Hz, and their levels in dB re the loudest:

$$\begin{aligned} &624.46, 1248.93, 1873.39, 2497.85, 3122.31 \\ &-1.56, 0.00, -3.23, -7.21, -11.98 \end{aligned}$$

The *second* harmonic is the loudest, the fundamental 1.56 dB below it. (The recording is Mihai Sorohan’s, from Freesound, CC BY 4.0, measured from librosa’s converted copy [32, 14]; the plot it rebuilds is the Hugging Face audio course’s [4].) One honest pointer: a real note lands between rows, so a window shaped those levels — the windowing chapter again.

The same ruler is everywhere. Decibels are $10 \times \log_{10}$ of a power ratio; pH is $-\log_{10}$ of hydrogen-ion activity [1]; stellar magnitude is $-2.5 \times \log_{10}$ of a brightness ratio, so that five magnitudes are a factor of 100 [23]; semitones are $12 \times \log_2$ of a frequency ratio, which makes a 3:2 fifth 7.0196 of them. Each is a constant times the log of a ratio — the logarithm chapter’s strip with its own stride. That the ear itself hears in logs is an approximate motivation, not a law [35].

1.7 The fold at half the sample rate

Why does the bank stop at row 4 — at 4000 Hz, half the sample rate? Because a row 5 would be row 3 again. At the 8 stops the 7-lap cosine probe *is* the 1-lap cosine probe, number for number, and the 7-lap sine probe is the 1-lap one with every sign flipped:

$$\cos(2\pi(8-k)n/8) = \cos(2\pi kn/8), \quad \sin(2\pi(8-k)n/8) = -\sin(2\pi kn/8).$$

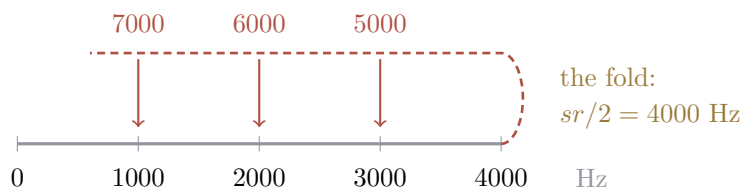
Likewise 6 laps is 2, and 5 laps is 3. Run the table of zeros past row 4 and it lights up: c_3 against c_5 adds to 4, s_3 against s_5 to -4 — not 0. A detector above row 4 would only repeat one below it.

The reason is on the first section’s circle. A 7000 Hz tone steps 7/8 of a lap between samples, visiting the angles (in degrees)

$$0, 315, 270, 225, 180, 135, 90, 45$$

— exactly the stops of a point stepping 1/8 of a lap *backward*. Same shadow, height flipped: it is the wagon wheel that seems to turn the wrong way [34]. Two cosines, 1000 Hz and 7000 Hz, pass through the same 8 samples, the samples cannot say which was played, and it is the 1000 Hz row that fires. On the frequency ruler, sampled at 8000 Hz:

$$5000 \rightarrow 3000, \quad 6000 \rightarrow 2000, \quad 7000 \rightarrow 1000.$$



(The ruler stops at 7000 on purpose. From 8000 Hz up the picture is no longer a mirror, and “cosine the same, sine flipped” stops being the rule.)

Now the first section’s promise is cashed. It said a tone needs at least two samples per lap; strictly, it needs *more than* two: it must lie *below* half the sample rate ($sr/2$), not at it [10]. Exactly at $sr/2$ a sine lands on 0 at every stop, and the samples are silence — the end row’s missing partner, met from the other side. (Textbooks disagree about what “the Nyquist frequency” names, so this chapter says “half the sample rate” instead [22].) And the impostor is a *clean lower tone*, not noise: nothing downstream can tell it from the real thing, so whatever lies above $sr/2$ has to be removed *before* sampling. Hence 8000 Hz for telephone speech and 16 000 Hz for speech models — where, by the same fold, $10\,000 \rightarrow 6000$ Hz. In numpy the fold is visible in the output itself: the upper half of `fft`’s output is this mirror, and `rfft` returns the bank. The sampling theorem’s credit line: E. T. Whittaker 1915, Kotelnikov 1933, Shannon 1949 [26, 15].

Where this goes. The series closes on a map. *Which tones are in it?* — the bank of detectors. *Quiet beside loud?* — decibels. Both are built. *Echoes and smoothing?* — convolution. *A tone between rows?* — windowing. *Tones that change over time?* — the short-time transform, whose columns stacked side by side are the spectrogram. *The ear’s own grouping?* — mel. Those are the road ahead, convolution first because leakage is best explained by it. Three further promises leave this chapter. The complex form — one complex number per row instead of a pair, and the reason the 28 zeros hold for every N — waits on Euler’s formula. The inverse, rebuilding the samples from the readings, is unbuilt. And the fast Fourier transform computes these same readings in far fewer steps by divide and conquer — an algorithms story, Cooley and Tukey’s in 1965 [2], though Gauss had the general formula around 1805, published in 1866 [3].

Practice problems

Problem 1.1. Eight samples are taken at 8000 per second. Here they are, above the 2000 Hz detector's two probes at the same 8 stops:

samples	5	12	-5	-12	5	12	-5	-12
cosine probe	1	0	-1	0	1	0	-1	0
sine probe	0	1	0	-1	0	1	0	-1

Compute the detector's pair (cosine sum, sine sum), its reading, and the amplitude of the tone; compare the amplitude with the largest sample. Then give the pair of the 1000 Hz detector, whose probes are $\cos(2\pi n/8)$ and $\sin(2\pi n/8)$ for $n = 0, \dots, 7$.

Hint: Each sum has only four nonzero terms. For the 1000 Hz detector, pair each sample with the one four places later.

Problem 1.2. All signals here are 8 samples at 8000 per second, read by the five-detector bank (0, 1000, 2000, 3000, 4000 Hz). (a) The signal is $2.5 \sin(2\pi \cdot 3000t + 50^\circ)$. Give all five readings. (b) The signal is $3 \sin(2\pi \cdot 1000t) + 0.5 \cos(2\pi \cdot 3000t)$. Give all five readings, and name the two facts that make the other rows read exactly 0.

Hint: On rows 1 to 3 a tone of amplitude A sitting on the row's frequency reads $4A$, whatever its starting angle.

Problem 1.3. For each bank give the window's duration, the spacing between detectors, the number of rows, and the highest row's frequency. (a) Whisper's: $sr = 16\,000$ Hz, $N = 400$. (b) $sr = 48\,000$ Hz, $N = 960$. (c) You need detectors 10 Hz apart at $sr = 16\,000$ Hz: how many samples, and how long a window? If you switch to $sr = 48\,000$ Hz and keep that N , what is the spacing — and what N restores 10 Hz?

Hint: Spacing = $sr/N = 1/\text{duration}$; rows = $N/2 + 1$; the top row is $sr/2$.

Problem 1.4. Use $\log_{10} 2 \approx 0.301$. Levels are in dB relative to the loudest row unless a power ratio is named. (a) The loudest row reads 4 and another reads 0.5: what is the second row's level? (b) A row sits at -12.04 dB: what is its amplitude ratio to the loudest row, and the power ratio? (c) What is the level of a tenth of the amplitude? Of a tenth of the power? (d) A tone has half the *power* of the reference: what is its level, and its amplitude ratio? (e) The loudest row reads 4; another row sits at -26.02 dB. What does it read?

Hint: $20 \log_{10}$ of an amplitude ratio, $10 \log_{10}$ of a power ratio; split every ratio into halvings and tenfoldings.

Problem 1.5. (a) A 5000 Hz tone is sampled at 8000 per second. At what frequency is it heard? (b) At 16 000 per second, where are tones of 5000, 9000 and 10 000 Hz heard? (c) Write the 8 samples of $\sin(2\pi \cdot 7000 t)$ taken at 8000 per second, starting at $t = 0$, to two decimals. Which 1000 Hz tone has exactly these samples? Give the 1000 Hz detector's pair (cosine sum, sine sum) and reading. (d) The same for $\cos(2\pi \cdot 7000 t)$: pair and reading on the 1000 Hz detector.

Hint: Between $sr/2$ and sr a tone lands at $sr - f$, its cosine part unchanged and its sine part negated.

Problem 1.6. On the $N = 8$, $sr = 8000$ Hz grid the top row listens at 4000 Hz, exactly $sr/2$. (a) Write the samples of $\cos(2\pi \cdot 4000 t)$. What does row 4 read, and what divisor recovers the amplitude 1? (b) Write the samples of $\sin(2\pi \cdot 4000 t)$. What does every row read? (c) Write the samples of $3 \cos(2\pi \cdot 4000 t + 60^\circ)$. What does row 4 read, and what amplitude does the end-row divisor report?

Hint: At 4 laps the sine probe is all zeros, so row 4 is its cosine sum alone; $\cos(\pi n + \varphi) = (-1)^n \cos \varphi$.

Problem 1.7. Exactly one of these five claims is true. Which — and what is wrong with each of the others? (a) “Between two samples a recording holds the earlier value, so the honest picture of 8 samples is a staircase.” (b) “8 samples at 8000 per second put the detectors 1000 Hz apart; 16 samples at 16 000 per second put them 500 Hz apart.” (c) “A row at 0 dB is silent.” (d) “A 7000 Hz tone sampled at 8000 per second shows up as noise spread across the rows.” (e) “Starting a 2000 Hz tone a quarter of a lap later changes the 2000 Hz detector's pair but not its reading.”

Hint: Check (b) with sr/N , (c) with $\log 1$ and $\log 0$, (d) with the fold.

Problem 1.8. [stretch] The tone of the first problem starts one sample later, so the samples read $-12, 5, 12, -5, -12, 5, 12, -5$; two samples later they read $-5, -12, 5, 12, \dots$; three samples later $12, -5, -12, 5, \dots$. Give the 2000 Hz detector's pair (cosine sum, sine sum) and its reading at each of the three delays, and say what the sine probe alone would have reported at delays 0 to 3. Through what angle does one sample's delay turn the pair — and through what angle would it turn a 1000 Hz tone's pair on the 1000 Hz detector?

Hint: The probes have not moved; only the samples have. A tone of f Hz advances f/sr of a lap per sample.

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